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REVIEW

Three strategies for claiming research progress

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Abstract

Researchers are increasingly challenged by this question: On what basis can they claim their work contributes to research progress? We propose that contemporary consumer research relies on three broad strategies for warranting such claims. An establishing-an-effect strategy claims progress based on confidence that a focal relationship between variables X and Y reliably replicates. A theory-of-the-effect strategy claims progress based on confidence that a local mechanistic account specifies when and how a particular effect occurs. A theoretical framework strategy instead claims progress based on confidence in an explanation that transcends any single effect paradigm. We (a) clarify how these strategies differ in their core logic, methodological emphases, and vulnerabilities; (b) document via a coding of recent articles in leading consumer research journals that the field relies overwhelmingly on the theory-of-the-effect strategy and rarely advances theoretical frameworks; and (c) develop actionable guidance for progressing from theories of effects to frameworks. We argue that recognizing the merits and vulnerabilities of each strategy—and deliberately leveraging the strategies in combination—provides a stronger basis for claiming research progress.

KEYWORDS

effects, philosophy of science, replication, theoretical frameworks

What should the end goal be in designing behavioral studies? How can a study claim progress in a given research area? There have been high-level discussions of issues relevant to these questions (e.g., Alba, 2012; Calder et al., 2019, 2021; Golder et al., 2022; Janiszewski & van Osselaer, 2021; Lynch et al., 2012, 2023). In terms of research practice, however, researchers presently pursue, at least implicitly, one of three very different general approaches to claiming research progress. The three strategies are as follows.

The *establishing-an-effect* strategy treats the reliable demonstration of a relationship between variables X and Y as the primary goal. The central question is whether a competent researcher, using similar procedures and adequate power, can expect to obtain the same effect. Claiming progress is grounded in confidence that an effect is replicable and robust, including in field or "real-world" settings.

The *theory-of-the-effect* strategy, which appears to be the dominant approach in consumer research, is based on a local mechanistic account of an effect. Researchers identify psychological constructs, specify mediating processes and moderating conditions, and test when and how the effect strengthens, attenuates, or reverses. Claims of research progress here are tied to conditional generalization: knowing under which theoretically specified conditions the effect will or will not occur.

The *theoretical framework* strategy shifts the basis for claiming progress further. Instead of explaining a particular effect paradigm, it seeks a set of constructs and theoretical relations that can account for why multiple, diverse phenomena, including both observed and not-yet-observed effects occur. Claims of progress are based in confidence in an explanation that transcends any individual effect to organize and make sense of diverse findings. In other words, the three strategies share the same

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overarching goal—warranting a claim that research has advanced knowledge—but depend on different bases for making a claim of progress.

Our goal in this article is not to argue for the superiority of one strategy over the others. Rather, we aim to (1) clarify what each strategy assumes, how it is typically implemented, and what kind of basis for claiming progress it supports; (2) highlight each strategy's characteristic vulnerabilities (generalization limits for effect-based work, theoretical shallowness and proliferation for theories-of-effects, and theory underdetermination for frameworks); and (3) provide guidance for how researchers can synergistically move between strategies.

The remainder of the article proceeds as follows. We first describe each strategy in more detail, including its core philosophy and methodological principles, and illustrate them with examples. We then evaluate what each strategy can and cannot deliver in terms of a basis for claiming cumulative progress. Next, we show how a phenomenon (e.g., the endowment effect, the attraction effect) can be pursued under each strategy, and we use recent theory-of-the-effect papers to illustrate how a framework can be used to connect and extend seemingly disparate findings. Finally, we draw out practical guidelines for researchers.

THE ESTABLISHING-AN-EFFECT STRATEGY

Research with the goal of determining whether an effect is real and robust falls into this category. An effect is established based on the premise that "any competent researcher should be able to obtain it when using the same procedures with adequate statistical power" (Simon, 2014, p. 76). For instance, investigations of the well-known endowment effect address whether individuals randomly assigned to own a product demand more to let it go than individuals are willing to give up to acquire it (Thaler, 1980). Claims of research progress depend on the extent to which studies using similar procedures and contexts to that of prior demonstrations can expect to replicate an effect.

Explanation is not required when applying this strategy. Rather, confidence is based on inductive generalization. As articulated plainly by Richard Thaler in a 2021 *Behavioral Scientist* interview, "Loss aversion is very real. Preference reversals exist. The ultimatum game, you play it anywhere in the world, if you offer 1 percent of the pie, you're going to get turned down. I won't go through all of them, but they all replicate and the effect sizes are all huge. That's why I picked them to write about 30 years ago, and nothing has changed… Now there are questions in the level of psychology about what causes that behavior. That's a question for psychologists. To an economist, I'm done. Buying and selling prices are different. That, I'm calling loss aversion."

Research practices

Researchers following this strategy design their studies to predict the effect of one variable on another where the variable relationship is thought to be significant as a phenomenon. According to this strategy, research ought to progress in three stages: (1) demonstrate-the-effect, (2) replicate-the-effect, and (3) generalize-the-effect. Researchers might also show "that their finding is replicable under multiple operationalizations of the key variables" (Simmons et al., 2021, p. 161). Different operationalizations of the same basic effect thereby define an "effect paradigm."

Rigorous demonstration and replication, especially if field study replications are conducted, are thought to provide confidence that the effect is not only replicable but also has external validity, meaning the effect can be generalized to other settings (Gal & Rucker, 2022). Indeed, there seems to be a growing expectation that articles contain a field study or other "real-world" data to demonstrate the real-world "generalizability" or external validity of the effect (Morales et al., 2017; Wang et al., 2015).

A focus on establishing effects is reflected in recommendations that call for conducting more independent replications of an effect, using large sample sizes, and requiring preregistration to avoid p-hacking (Simmons et al., 2021), postdiction (Nosek et al., 2018), and publication bias (Simonsohn et al., 2014). This strategy seeks validity through the rigorous demonstration and replication of effects, ideally registered in advance. Conversely, failures to replicate identify effects that are untrustworthy by virtue of questionable research practices (John et al., 2012), methodological errors (Simmons et al., 2021), or chance results (Nosek et al., 2018). In short, successful replications establish confidence in results by verifying effects and failures to replicate identify invalid effects.

At times, in the context of establishing an effect, boundary conditions for the effect are identified, but in a largely atheoretical way. For instance, in the classic endowment effect demonstration, randomly assigned owners of an object are found to demand more to part with it than randomly assigned nonowners are willing to pay to acquire it (Kahneman et al., 1990). Yet, an endowment effect in everyday transactions in which buyers are not willing to pay what sellers demand would be reflected in severe impediments to commerce. To address this issue, Tversky and Kahneman (1991) proposed a boundary condition on the endowment effect, namely that it does not apply to routine commercial transactions. Similarly, when it comes to replication, failures to replicate might be construed not as undermining the effect, but as identifying boundary conditions (Nosek & Errington, 2020). Even with attention to boundary conditions, the overriding aim is to establish the effect by robust demonstration and replication.

There are contentions that failures of effects to replicate in new settings (Bohner & Schluter, 2014) or to establish the boundary conditions under which effects will or will not occur (Almaatouq et al., 2024; Yarkoni, 2022) should raise concerns about a "reproducibility crisis" and "generalizability crisis" respectively. These contentions have reinforced the view that effects should be the end goal of research. To be clear, what distinguishes this view is not just that empirical findings are crucial in the research process, but also that conclusions about replicability of effects are the ultimate goal that justifies confidence in research.

THE THEORY-OF-THE-EFFECT STRATEGY

From our reading of consumer research (confirmed by our coding of recent papers, more on this below), the theory of the effect is the most common strategy for claiming research progress. The appeal is that it focuses on effects while at the same time introducing an element of theory. Studies adopting this strategy go beyond a determination of whether an effect of interest occurs and replicates reliably, given sufficient methodological rigor. And they go beyond identifying boundary conditions in a mostly atheoretical way.

Adopting the theory of the effect strategy prompts a focus on identifying the mechanism or process underlying an effect of interest in order to understand key conditions under which it will or will not occur, and thereby to generalize it across diverse settings (i.e., external validity). Theory takes the form of associating variables with more abstract ideas or concepts ("constructs"). Support for the theory tends to come from multiple operationalizations of key constructs, e.g., a variable entailing a speaker with academic credentials and another variable entailing practical experience both in theory become source credibility. Support can also come from identifying how the effect occurs (i.e., a mediating mechanism) and specifying theoretically motivated conditions under which the effect does and does not occur (i.e., moderators).

Research practices

In a prototypical example of this strategy, Gal and Rucker (2012) describe what they term a "response substitution" effect in surveys, where respondents to a closed-end survey question attempt to convey something they wish to express in response to an unrelated question rather than answer what the question is actually asking about. For instance, respondents might reply to a question about the atmosphere of a restaurant negatively if the service is bad even if they understood the question was about the atmosphere and had a favorable impression of it. In other words, they attempt to convey their negative view of the service in response to the question about the atmosphere. The authors theorize that this effect occurs due to respondents' need for self-expression. They develop this theorizing by integrating past research suggesting people express themselves through their choices with research indicating that attitudes can have self-expressive functions.

Gal and Rucker report three studies to document the response substitution effect and test their theorizing using the same basic procedure across studies: respondents are primed by a scenario to adopt a particular attitude and then asked an unrelated question about the scenario. Response substitution is reflected in the degree to which the attitude participants are primed to adopt influences their response to the unrelated question. For example, in one study the opinion of an actor's intelligence is solicited after the presentation of a scenario where an actor was or was not wasteful, and in another the opinion of product quality is solicited following a scenario where the product manufacturer was or was not engaged in immoral practices. Besides using different operationalizations of variables across studies, the authors also use moderators to support their theorizing. They show that those given an opportunity to express the attitude they wish to express prior to being asked the focal (unrelated) question are less likely to engage in response substitution (experiments 1–3). By contrast, those who view the attitude they wish to express as relatively more important are more likely to engage in response substitution (experiment 2).

As is reflected in this example, research articles following the theory-of-the-effect strategy tend to contain multiple studies with different operationalizations and moderators of the same basic effect paradigm. In other words, similar to the establishing-an-effect strategy, the teleology of the theory-of-the-effect strategy is on establishing the effect, albeit paired with specifying its mechanism. This focus on the effect is reflected in an emphasis on knowledge accumulation in terms of effects, including identifying an "effect size" and key moderators of effects through meta-analysis (McShane & Böckenholt, 2017). For instance, with respect to studies on loss aversion effects, it has been argued through meta-analysis that the "coefficient of loss aversion" is about 2, with a higher coefficient in field experiments and in the general population versus the university population (Brown et al., 2024).

In some investigations that apply the theory-of-the-effect strategy, not all of the studies are instantiations of the same basic effect paradigm. In these cases, the underlying process hypothesized for an effect may be tested by reporting converging effects (Spencer et al., 2005). Nonetheless, the focus remains on demonstrating and explaining the effect. For example, Brough et al. (2016) show that men are less likely than women to engage in green behaviors, an effect they attribute (a) to helping behaviors, including helping the environment, being more stereotypically feminine, and (b) to the gender identity threat men perceive in engaging in stereotypically feminine behaviors.

Several studies are reported to test this theorizing. In one study, an Implicit Association Test (IAT) is administered to document an association between green behaviors and femininity. In other studies, it is shown that an actor in a vignette is perceived to be more feminine the more she/he cares for the environment (e.g., he/she chooses to use a reusable grocery bag). And in still other studies the fear of being perceived as feminine influences the choice options of men associated with caring for the environment (e.g., choice to donate to environmental charity). While these studies are not all instantiations of the same basic effect paradigm, they all converge on attempting to identify the mechanism underlying the effect that men are less likely than women to engage in green behaviors. Through identifying this mechanism, the authors' aim is to predict when the effect will or will not occur—when green behaviors imply a lack of masculinity.

In sum, as these examples indicate, applying the theory-of-the-effect strategy identifies when and how effects occur by the introduction of moderators and mediators implied by a mechanistic explanation. Although studies following a theory-of-the-effect strategy use theory, research progress is justified primarily by showing that an effect occurs and by identifying moderating variables. In other words, claims of progress are identified with external validity, the concept that effects can be generalized to new settings (Calder et al., 1982). Through developing an explanation that identifies the relevant moderators of an effect, the contention is that we can know in which settings the effect will or will not occur. Although it is certainly possible for such studies to stimulate advances in theory, these studies tend to focus on localized explanations and do not claim progress through contributing to a general explanation of a broader research area.

THE THEORETICAL FRAMEWORK STRATEGY

While the terms theory and theoretical framework are used in diverse ways, here we use the term theoretical framework in distinction to a theory of the effect. A theoretical framework is any theory that does not depend on any one observed effect or set of related effects (i.e., an effect paradigm). Rather it provides an explanation for *why* a variety of observed and not yet observed effects occur.

It is worth noting that some mechanisms, such as certain dual-process models (System 1 vs. System 2), are sufficiently abstract to explain when and how effects occur across multiple effect paradigms and thereby to figure in different theories of effects. However, such mechanisms do not by themselves explain why people activate one process rather than another in a given situation. That is, these are portable mechanisms (useful building blocks that can be embedded in many theories of effects or frameworks), but they are not themselves frameworks unless they specify the conditions and higher-order constructs that determine when the mechanism is engaged. Conversely, a theoretical framework provides an account of *why* diverse effects occur. Examples of theoretical frameworks range from cognitive dissonance theory to reason-based choice to evolutionary psychology frameworks.

The theoretical frameworks strategy recognizes the crucial role of documenting whether an effect can be replicated as well as identifying effect mechanisms. However, claims of research and scientific progress are defined by how well a theoretical framework explains why diverse effects occur (Muthukrishna & Henrich, 2019). Thus, theoretical frameworks revolve around problems of explanation of behavior broadly construed (Calder et al., 2021; Isaac & Calder, 2024).

The theoretical frameworks strategy makes use of studies associated with the other two strategies but it does not assume that these strategies are sufficient to justify claims of research progress. Again, the point is that a theoretical framework provides greater explanatory power—a level of explanation that moves beyond a specific effect and in doing so justifies claiming research progress. Both effects and theory are important but claims of progress are grounded in the growth of theoretical frameworks. Enhancing the breadth of phenomena that can be explained by a theoretical framework enhances the confidence in the inferences derived from it.

The distinction between the theory-of-the-effect and theoretical frameworks strategies can be analogized to the distinction in levels of explanation hypothesized for animal behavior (Mayr, 1961; Tinbergen, 1963). A lower level of explanation, sometimes labeled a proximate cause or explanation, focuses on the immediate mechanism through which an animal engages in behavior (e.g., the underlying physiological, neurological, and hormonal mechanisms that produce a behavior); a higher level of explanation, sometimes labeled an ultimate cause or explanation, focuses on the broader explanation for behavior, such as the behavior's evolutionary function (e.g., how does a behavior's function increase an animal's chance of survival and reproduction?). Although both entail explanation, the former is more akin to explaining "how" a behavior occurs and the latter more akin to explaining "why" it occurs at all. By asking both "how" and "why," scientists can gain a more complete and nuanced understanding of animal behavior.

Likewise, theories of effects examine proximate causes that explain immediate mechanisms for specific effects, while theoretical frameworks provide the broader context to explain why a particular sort of behavior exists in the first place and thereby transcend specific effects or specific effect paradigms. These two levels of analysis are not mutually exclusive; they are interconnected, as will be discussed later.

Research practices

Addressing the question of why diverse effects occur entails presenting an explanation at a level of abstraction that transcends individual effects or effect paradigms. Like a theory of the effect, a theoretical framework introduces constructs. However, unlike a theory-of-the-effect strategy, the constructs are not tied to variables from any single effect paradigm. This is not to say that theories-of-effects do not employ constructs that transcend an effect paradigm, such as cognitive load or accessibility—they do. However, these constructs are used in the service of developing a specific, local mechanistic explanation for a specific effect or effect paradigm.

Articles employing a theoretical framework strategy, by contrast, aim to develop, support, or extend the scope of a theoretical explanation for diverse effects. As a recent example, He and Sternthal (2023) developed an integrative theoretical framework, labeled the “3A framework,” to explain how individuals allocate cognitive resources in judgment and decision-making. The 3A framework introduces three theoretical constructs—ambiguity, adoptability, and accessibility.[1] These constructs occur in the literature (e.g., Bettman et al., 1998; Higgins & Brendl, 1995), but they were not previously integrated as part of a framework.

Ambiguity arises in situations where the available information for processing is associated with a lack of clarity about the priority of one goal over another (e.g., effort conservation versus accuracy enhancement). Adoptability refers to which goal is most salient among alternative effort goals, which influences the choice between the alternative effort allocation goals. Accessibility refers to information processing that occurs in response to the adopted goal. Though both adoptability and accessibility can be influenced by the salience of information, adoptability relates to the relative salience of goals whereas accessibility is about the ease with which information is retrieved and processed. For example, an effort conservation goal prompts automatic processing, whereas an accuracy enhancement goal entails deliberative processing of internal and stimulus information (Kahneman, 2011), often in comparison to some referent (Malaviya, 2007; Meyers-Levy, 1991). In many cases, the goal that is adopted guides the information that is accessible in making a judgment. However, there are situations where the accessibility of information overrides or guides goal adoption in the judgment process (Higgins & Brendl, 1995).

As is evident from the description, neither the constructs that make up the 3A framework nor the linkages among them in the framework are specific to any particular effect paradigm but apply broadly to cognitive resource allocation. He and Sternthal (2023) developed the framework to explain both demonstrations and failures to replicate the attraction effect (a similar but inferior decoy affects choice). They make sense of past failures to replicate it in situations with different stimuli or contexts than in the original demonstrations. What differentiates their article from attraction effect studies employing a theory-of-the-effect strategy is that the objective is not predominantly to explain the mechanism for and thereby the moderators of the attraction effect, but rather to develop a theoretical framework. The framework's ability to explain the attraction effect and reconcile failures to replicate it is predominantly used to support and illustrate the explanatory power of the theoretical framework.

Other examples of research employing the theoretical framework strategy focus on supporting the explanatory power and scope of an evolutionary framework of consumer behavior (e.g., Griskevicius et al., 2010) and supporting the explanatory power and scope of the elaboration likelihood model to explain effects of misinformation (Petty, 2024). Yet other research employing this strategy develops a reason-based framework whereby people's decisions are influenced by the need to justify them, whether explicitly or implicitly, and whether to themselves or to others (Shafir et al., 1993). Shafir et al. develop the framework and then illustrate its relevance through a number of experiments involving different effect paradigms and then describe how it can account for other diverse effect paradigms in the literature.

Once a relevant framework is identified, the scope and parsimony of its construct-to-construct relations in explaining effects can be determined, preferably in comparison to rival theories, to explain a range of observed effects. Calder et al. (2023) illustrate the scope and parsimony of the 3A theoretical framework by applying it to explain the repetition effect (presenting people with more information or with greater frequency affects persuasion), the depletion effect (a high effort initial task affects performance on a subsequent unrelated task), and the number of exemplars effect (considering more beliefs or arguments related to an attitude affects subsequent favorableness). In sum, theoretical frameworks transcend individual effect paradigms through linking constructs at a level of abstraction that is not tied to a particular effect paradigm to explain why effects occur.

To facilitate appreciation of the distinctiveness of the three strategies for claiming progress discussed in this

[1] He and Sternthal (2023) use the term “applicability” when introducing the 3A framework. Calder et al. (2023) replace “applicability” with “adoptability.” However, these refer to the same concept.

TABLE 1 Central tenets of three strategies for claiming research progress.

(a) Core philosophy

Establishing an effect strategy	Theory-of-effects strategy	Theoretical framework strategy
Effects as endpoints	Effects plus mechanisms	Explanation transcending effects
The strategy treats demonstrable, replicable effects (relationships between variables X and Y) as the ultimate goal and justification for claiming research progress, rather than as means to understanding underlying mechanisms	The strategy combines demonstrating effects with identifying the underlying psychological mechanism or process that produces them. The goal is understanding not just *that* an effect occurs, but *how* and *when* it occurs	The ultimate goal is developing theoretical explanations sufficiently abstract to transcend any particular effect or effect paradigm. Justification for claiming research progress comes from explanatory power across diverse phenomena, not from establishing individual effects
Inductive generalization over explanation	Conditional generalization	Why over how
Confidence that an effect will replicate is built through repeated empirical demonstration rather than theoretical explanation. Once an effect reliably replicates, the research is considered "done" regardless of whether the underlying psychological mechanisms are understood	Confidence that an effect can be generalized to diverse settings comes from specifying the theoretical conditions under which effects will or will not occur, enabling prediction across diverse settings rather than just procedural replication	While theories-of-effects explain proximate causes (the immediate mechanism for how specific effects occur), theoretical frameworks explain ultimate causes (why particular sorts of behavior exist in the first place). This parallels the distinction in animal behavior research between proximate and ultimate explanations
Use of field studies	Localized theoretical explanations	Growth of frameworks lead to progress
Demonstrating effects in real-world settings or field studies is increasingly expected to show generalizability beyond laboratory contexts. The assumption is that demonstrating an effect in a field study shows that it applies to diverse and real-world settings	Theory is invoked to explain specific effects through mediating mechanisms and moderating conditions, rather than to build broad, cross effect explanations of why effects occur	Claims of scientific progress is grounded in how well a theoretical framework explains why diverse effects occur, not by accumulating documented effects or their moderators

(b) Methodological principles

Establishing an effect strategy	Theory of effects strategy	Theoretical framework strategy
Three-stage research progression	Construct-based theorizing	Theorizing across effect paradigms
Research should advance through (1) demonstrate-the-effect, (2) replicate-the-effect, and (3) generalize-the-effect	Variables are associated with more abstract concepts or "constructs" (e.g., different operationalizations all representing source credibility). Studies demonstrate the same basic effect but claims of research progress are justified by common constructs rather than pure replication	Frameworks introduce constructs that are not tied to variables from any single effect paradigm. Unlike theories of effects that use constructs to explain specific local mechanisms, frameworks employ constructs that apply broadly across multiple effect paradigms (often by specifying why such mechanisms are activated, rather than only showing how an effect works in a given paradigm). The focus is on developing relationships between abstract theoretical constructs that can generate predictions for diverse effects
Replicability as the progress criterion	Mediation and moderation testing	Theoretical integration
An effect is considered "real and robust" when any competent researcher can obtain it using the same procedures with adequate statistical power. Successful replications validate effects; failures to replicate invalidate effects. Empirical replicability itself provides sufficient warrant for claiming research progress, making theoretical explanation secondary or unnecessary	Research tests predictions about (1) the process by which effects occur (mediators) and (2) theoretically motivated conditions that strengthen or weaken effects (moderators)	Frameworks often integrate existing theoretical concepts in novel ways rather than necessarily introducing entirely new constructs, creating more comprehensive explanatory systems rather than the proliferation of different constructs

TABLE 1 (Continued)

(b) Methodological principles

Establishing an effect strategy	Theory of effects strategy	Theoretical framework strategy
Procedural standardization	External validity through theoretical specification	Explanatory scope and parsimony
Progress is claimed based on confidence that studies using similar procedures and contexts can expect to replicate the effect. The focus is on procedural consistency. Demonstrating effects in real-world settings or field studies is increasingly expected to show generalizability beyond laboratory contexts	Generalizability is shown by identifying which theoretical conditions enable prediction of when effects will transfer to new settings	Frameworks are evaluated by their ability to explain a range of observed effects with fewer ancillary or ad hoc assumptions than rival theories. Greater breadth of phenomena explained enhances claims of research progress

section, the main tenets of each strategy are summarized and encapsulated in Table 1.

PREVALENCE OF THE STRATEGIES IN CONTEMPORARY CONSUMER RESEARCH

To evaluate the prevalence of the different strategies, we categorized the most recent year of articles from each of the *Journal of Consumer Psychology* (JCP) and the *Journal of Consumer Research* (JCR) (from July 2025 through April 2026 for JCP and from June 2025 to April 2026 for JCR; see Appendix S1 for details). We excluded articles that were purely conceptual or purely qualitative. We focused on empirical papers because our interest is in how researchers claim research progress through the combination of theoretical argument and empirical evidence. This yielded 66 articles in total, 41 from JCR and 25 from JCP.

We categorized articles based on assessing the match with the criteria in Table 1. Besides, the three strategies in the table, we also categorized articles into a predominantly methodological type of article. We performed the analysis ourselves by evaluating how the article's theory and methods aligned with the criteria in Table 1. In doing so, we were aided by ChatGPT 5.5 (extended thinking). In particular, we prompted ChatGPT to provide us with summaries of articles' theorizing and methods to facilitate our categorization.

Overall, we categorized 24 of the 25 JCP articles as following a theory-of-the-effect strategy and one (1) as following an establishing an effect strategy. Of the 41 JCR articles, we categorized 36 as following a theory-of-the-effect strategy, five (5) as following a methodological strategy, and none as following an establishing an effect strategy. As a basis of comparison, and applying the same criteria, we categorized Gal (2006), Shafir et al. (1993), Griskevicius et al. (2010), and He and Sternthal (2023) as relying on the theoretical framework strategy (see Appendix S1).

EVALUATING THE ALTERNATIVE STRATEGIES

We now turn from describing the three strategies for claiming research progress to evaluating what each strategy can and cannot deliver. Because all three strategies are visible in current practice, it might be tempting to treat them as interchangeable routes to the same goal. Our view is that they differ in how they support cumulative progress.

The establishing-an-effect strategy

The establishing-an-effect strategy warrants claims of research progress in the empirical demonstration of a focal relationship. Although our survey of the consumer literature indicates that the establishing-an-effect strategy is less common today, we think that it is widespread in the broader behavioral literature. Also, as exemplified in the interview with Richard Thaler, it is a view that has received considerable advocacy.

Replication is key to this strategy. The conception is that if multiple, well-powered studies, ideally including field experiments, keep showing that X affects Y, then researchers can be confident that the effect is "real." What distinguishes this view is not just that empirical findings are crucial in the research process, but also that conclusions about the occurrence of "effects" are the ultimate goal that justifies claims of research progress.

Replications are, of course, useful. All else being equal, more data are better than less, and replication attempts can identify methodological flaws in original work. Large-scale initiatives have appropriately raised concerns about false positives, undisclosed flexibility in analysis, and publication bias (e.g., Ioannidis, 2005; John et al., 2012; Simmons et al., 2011). Requirements such as preregistration, higher statistical power, and independent replication are sensible safeguards against p-hacking and other questionable research practices (Wedel & Gal, 2024).

At the same time, treating the demonstration of an effect and its replication as the primary or sufficient criteria for claiming progress is problematic. First, replications are never literally exact (Urminsky & Dietvorst, 2024). They necessarily differ in time, population, context, or seemingly minor procedural details. As a result, failures to replicate can arise from contextual differences rather than from the absence of the underlying effect (Shiffrin et al., 2017; Stroebe & Strack, 2014; van Bavel et al., 2016). Even when a study is advertised as a "close" or "direct" replication, there are no practical guidelines for how much contextual variation is acceptable or for how to weight conflicting findings across studies.

Research on nudging interventions to increase vaccination illustrates these difficulties. A systematic review of nudges for vaccination concluded that the effectiveness and even the direction of effects vary substantially across settings, with mixed evidence for many approaches and a clear need for context-sensitive adaptation (Reñosa et al., 2021). Mega-studies, in which many interventions are tested simultaneously in one population (Milkman et al., 2021), address some concerns about contextual variation within that population. For example, Milkman et al. (2022) tested 22 different text messages among nearly 690,000 Walmart customers and identified a top-performing message ("a vaccine is waiting for you" sent twice, 3 days apart) that was not anticipated by expert forecasters. Yet even here, nothing guarantees that the same messages will be effective at a different time, in a different country, or for a different disease. Indeed, another study found that interventions effective for increasing flu vaccination did not carry over to COVID-19 vaccination (Jacobson et al., 2022).

Meta-analysis is sometimes proposed as a complementary remedy: aggregate the available studies, assess the average effect, and estimate its moderators. But meta-analytic conclusions can themselves be challenged by later work. For instance, a meta-analysis of accuracy prompts concluded that they constitute a "replicable and generalizable" intervention to reduce misinformation sharing (Pennycook & Rand, 2022), yet subsequent replications in some populations have failed to find the effect (e.g., Pretus et al., 2021). Meta-analysis is also vulnerable to the same contextual sensitivity and model-specification issues that plague individual studies.

Some scholars therefore argue that the relationship between original demonstrations and replications should be understood in statistical, not categorical, terms (e.g., Schwarz & Strack, 2014; van Aert & van Assen, 2018). On this view, each study is a noisy draw from a distribution of possible outcomes (Patil et al., 2016). Failures to replicate must then be "weighed" against prior evidence rather than treated as decisive. But this position offers no principled rule for how that weighing should be done, nor does it explain why failures to replicate should necessarily count against prior findings when the failures may themselves reflect poor contextual fit (van Bavel et al., 2016).

Consider a field experiment conducted in the United States to test the effectiveness of different communications on hotel guests' reuse of bathroom towels (Goldstein et al., 2008). The authors found that nudges indicating that the majority of guests reuse their towels were more effective in prompting towel reuse than ones describing the environmental benefits of this practice. This nudge failed replication in a subsequent investigation conducted in Germany (Bohner & Schluter, 2014). Whether the replication failed or succeeded, however, could not compromise the original effect because of their differences in time and setting. It is not that replication is undesirable; as we stated earlier, more studies of an effect are always better than fewer. The issue is that failures to replicate do not in and of themselves reveal which studies should be rejected and which should not.

In sum, the establishing-an-effect strategy puts faith in rigorous methods and replication as the basis for having confidence that a certain effect exists, and thereby as the basis for claiming research progress. This assumption seems so normal, in the sense of Kuhn (1970), that the strategy may be accepted as natural and anodyne. No doubt any failure to replicate is relevant to whether an empirical result should be questioned, but the establishing-an-effect strategy depends on the assumption that some, if not many, effects will be found to replicate. Examination of effects such as various nudging effects, however, raises the possibility that supposedly replicated effects may turn out to be contingent (van Bavel et al., 2016). In the end, claiming research progress by establishing an effect through replication is a strategy that should be considered on its own terms, not as an authoritative prescription.

In essence, it cannot be assumed that a reported effect will be replicated. Taking into consideration the issues we identified, a researcher employing this strategy should be able to give specific reasons for claiming that a reported effect will be replicable in future research. It should not merely be assumed that it will be unless shown otherwise.

The theory-of-the-effect strategy

The theory-of-the-effect strategy adds a layer of mechanistic explanation. This entails documenting that X affects Y, hypothesizing a process by which this effect occurs, testing this process (typically via mediators and moderators), and using it to forecast when the effect will be strengthened, attenuated, or reversed. This strategy is attractive because it promises both explanation and improved external validity: if we know how an effect works, we are in a better position to identify the conditions under which it will or will not occur.

A limitation of the strategy is that it remains tied to a particular effect paradigm and to the demonstration of empirical findings about when effects will or will not occur. This is a limitation because no mechanism fully specifies all the circumstances under which an effect will appear or disappear. Any given effect can be determined by multiple different mechanisms, and it can likewise be attenuated or eliminated by multiple different mechanisms. Furthermore, a novel setting can introduce moderators that make a previously identified mechanism moot or that make previously identified moderators yield different effects. As such, theory-of-the-effect papers are limited to finding specific process explanations and identifying moderators for effects within the conditions studied.

This limitation cannot be "fixed" simply by adding more moderators or mediation paths. Settings always differ from each other, and there is an open-ended set of potential moderators and mechanisms. This severely constrains the ability to generalize an effect. Yarkoni (2022) and others have, in fact, argued that we are facing a "generalizability crisis": our theories are only supported within the narrow contexts in which they were tested (see also Almaatouq et al., 2024).

The generation of theories that are localized to effects also leads to concerns that we are not making research progress (Gergen, 1978; Hutmacher & Franz, 2025). Proliferating endless local theories to account for specific effects risks constraining progress at a higher level. We can end up accumulating one-off theories of effects, rather than converging on broader, unifying explanations.

Theory-of-the-effect papers are typically organized around a focal effect: which variables moderate it through which mediators, and under what conditions it disappears or reverses. This emphasis reflects a natural desire to make effects more plausible and to show that they are generalizable. But it also means that progress is often equated with elaborating the mechanism and moderators of a particular effect rather than with explaining a wider class of phenomena.

The theory-of-the-effect strategy moves one step up from the simpler Establishing an Effect strategy by adding mechanistic elaboration. Yet the claim to research progress it supports remains local: it tells us when and how a particular effect is likely to emerge, not why a broader family of behaviors exists or how that family relates to other phenomena.

The theoretical framework strategy

The theoretical framework strategy shifts the focus from explaining specific effects to explaining why a range of different effects occur. Frameworks seek a higher level of abstraction, assembling constructs and relations that can account for multiple, diverse effect paradigms. This strategy aims to organize and explain an array of effects.

The promise of theoretical frameworks is increased explanatory power, scope, and parsimony. A good framework shows why effects that look different on the surface can be understood as arising from the same underlying explanation; it also helps anticipate when new effects should arise. In practice, these benefits are often realized by integrating existing constructs in new ways. The 3A framework, for example, brings together ambiguity, adoptability, and accessibility—constructs that appear in prior work on information processing and effort allocation—but links them into a system designed to explain how people allocate cognitive resources across tasks (He & Sternthal, 2023).

At the same time, it should be recognized that frameworks face a classic philosophical challenge: the underdetermination of theory by data (Stanford, 2021). For any finite body of findings, multiple plausible frameworks can often be constructed. Because different phenomena may be amenable to different theoretical treatments, there is no guarantee that a single framework is uniquely justified by the available data. There is also the danger that in developing frameworks with the scope to explain multiple effects, theory can become overly general or fitted ad hoc to account for different effects. In practice, this means that frameworks must be evaluated comparatively, on criteria such as explanatory power, scope, parsimony, coherence, and the ability to generate new, discriminating predictions that are empirically supported (Mackonis, 2013; Thagard, 1978) (Though note that there is little agreement on how to weight different criteria).

The evaluation of a theoretical framework should not depend on when the explanation is developed. Certainly, a framework explanation should not be treated as established merely because it can be fitted to a set of findings after the fact. But neither is post hoc timing, by itself, a reason to disregard the explanation. Frameworks in all sciences are often evaluated retrospectively as well as prospectively: what matters is whether the framework increases explanatory integration, reduces ad hoc assumptions relative to alternatives, and generates predictions that can later discriminate among competing accounts.

As indicated previously, the relationship between theories of the effect and theoretical frameworks is analogous to the distinction between proximate and ultimate explanations in animal behavior (Mayr, 1961; Tinbergen, 1963). Theories of effects typically address proximate "how" questions: by what immediate processes does X produce Y in this paradigm? Frameworks address more ultimate "why" questions: why do decision-makers exhibit this class of behaviors at all, and why does it take the observed form across varied situations? The two levels are thereby complementary and interconnected. Yet most contemporary consumer research remains focused on proximate explanations, leaving more expansive explanations underdeveloped.

From this evaluation, our conclusion is not that establishing an effect or theories-of-the-effect are without merit. Rather, it is that a reliance solely on these strategies does not take advantage of robust frameworks that organize and explain a broad range of phenomena as an additional basis for warranting research progress.

As we have seen, all three strategies in fact have vulnerabilities that pose challenges for claiming progress based on that strategy alone. Table 2 provides a summary of the challenges facing each strategy.

SAME PHENOMENON, DIFFERENT STRATEGIES

The preceding sections have described and evaluated the three strategies by drawing on different studies from different research areas. This corresponds with the current practice of researchers typically adopting one strategy. Nonetheless, it is instructive to consider cases in which the same empirical regularities have been approached over time with different standards for claiming research progress. Besides further illustrating how the choice of strategy shapes research design, these cases suggest how the three strategies could be complementary.

Endowment effect

We can illustrate how the same phenomenon can be pursued with all three strategies using the endowment effect. Knetsch's (1989) classic work is prototypical of an *establishing-an-effect* strategy. Across a series of simple trading experiments (e.g., mug vs. money, mug vs. chocolate bar), he shows that randomly endowed owners reliably demand far more to be willing to accept (WTA) an offer for an item than nonowners are willing to pay (WTP) to acquire it, while ruling out wealth effects, transaction costs, and strategic misreporting.

The contribution is not an explanation, but an inductive claim: any competent researcher who randomly assigns ownership and uses similar procedures should observe a large and robust WTA–WTP gap. Confidence that the effect can be replicated comes from repeatedly demonstrating that the effect replicates across goods, designs, and samples. Nonetheless, the limitation of the strategy is also apparent, as multiple documented attempts have failed to replicate the endowment effect (Humphrey et al., 2017; McGranaghan & Otto, 2022; Ziano et al., 2020; Ziano & Villanova, 2021; for review, see Zeiler, 2018).

Subsequent work on the endowment effect shifts to a *theory-of-the-effect* strategy by proposing specific psychological processes and testing their implications. Peck and Shu (2009), for example, argue that mere touch increases psychological ownership, which in turn produces endowment-like valuation asymmetries. Weaver and Frederick (2012) propose a reference-price account in which buying and selling prices are constructed relative to different salient price standards. Strahilevitz and Loewenstein (1998) focus on ownership history as a driver of attachment and anticipated regret. And Nayakankuppam and Mishra (2005) show that buyers and sellers literally *think about different features* of the same object—sellers retrieve more positive and fewer negative attributes than buyers—thereby producing divergent valuations.

Each of these research programs treats the endowment effect itself as the focal phenomenon and develops a process-level explanation that yields specific predictions about mediators (e.g., psychological ownership, feature retrieval) and moderators (e.g., touch, usage, elicitation procedures, experience with the object). Claims of progress here derive from how well the proposed mechanism accounts for when the endowment effect appears, disappears, or reverses, rather than from additional demonstrations of the effect per se. The limitation of this strategy is also evident as none of these accounts specify all the conditions under which the endowment effect appears or disappears (Gal & Rucker, 2018).

By contrast, Gal's (2006, 2021) omission–commission (action–inaction) account exemplifies a *theoretical framework* strategy. Rather than offering another endowment-specific mechanism, he argues that many instances of the endowment effect can be understood as a special case of a broader tendency to favor omissions over commissions. On this view, retaining an endowed object is coded as an omission, whereas trading it away is coded as a commission; because people interpret commissions as more causal than omissions, they are more averse to harmful commissions than to equally harmful omissions, and they therefore demand more to act (sell) than to refrain (keep).

This framework is more abstract than a theory-of-the-effect: it is meant to organize not only endowment phenomena but also status-quo bias, avoidance of (certain) risky bets, and others. As a result, it also makes predictions for different effects and effect paradigms. For instance, it predicts when people will and will not accept risky bets, and that willingness to pay to retrieve a lost object (losing it through omission) will be lower than willingness to accept to give up the same object (losing it through commission) (Gal & Rucker, 2018). Note that this framework need not explain all instances and moderators of the endowment effect, but only those that can be mapped onto omission versus commission. This is a reflection of its being pitched as a more general explanation of behavior than an explanation of the endowment effect per se.

The 3A framework of He and Sternthal (2023) provides another, distinct theoretical framework that can

TABLE 2 Challenges for strategies in claiming research progress.

Establishing-an-effect strategy—Generalization issues	Theory-of-the-effect strategy—Theoretical-depth issues	Theoretical framework strategy—Underdetermination issues
The replication paradox	Surface-level constructs	Too many theories problem
Replications can never be exact because they necessarily differ in time and setting from the original study. Thus, context can always affect whether an effect replicates. The strategy assumes many effects will prove replicable, but effects may turn out to be more contextually contingent than is presumed. Moderator testing cannot fully exhaust the potential contextual factors that might influence an effect	This strategy often identifies explanations that are close to independent variables—essentially describing the effect rather than genuinely explaining it. The strategy risks generating numerous disconnected, study-specific explanations. Moreover, typically many alternative local theories are possible. Any effects can be attributed to different local theories and can be eliminated by various factors. No single theory can uniquely account for an effect, which means local explanations are necessarily incomplete	The number of possible theories is unlimited because theoretical frameworks are inferential. New theories can always be constructed by researchers. Theories are therefore inherently underdetermined by existing data. A framework may thus be only one of many that could fit the data
The attribution problem	Plausibility instead of explanation	The problem of criteria
Failures to replicate do not reveal which studies should be accepted or rejected. Such failures could result from the original study being flawed or from the replication lacking contextual sensitivity to crucial variables	Theory can be intended to make the effect seem more plausible and believable. It creates a narrative bridge between the independent and dependent variables that feels intuitively reasonable. This sense of plausibility should not be confused with true explanatory depth. The explanation remains local—tailored specifically to make sense of the particular effect observed in that study	Criteria such as explanatory power, scope, falsifiability, and parsimony can serve as criteria for considering whether a theoretical framework is the best explanation, but other criteria are possible. How to weight different criteria or resolve discrepancies may be difficult
The weight problem	The problem with mediation	Explaining too much
There is no systematic way to determine how much weight a replication failure should carry or how many successful versus failed replications are needed to establish confidence. Each replication remains open to whether further replication is needed	Mediating variables are often taken as representing theoretical constructs rather than as variables. However, they may not really explain *why* the mediating process occurs. In fact, mediators may be alternative measures of dependent variables and thus correlate with them	Developing theoretical frameworks with the scope to explain many effects risks developing theories that can explain anything. Theoretical frameworks must maintain separation between theoretical arguments and evidence while allowing them to mutually inform each other

also encompass the endowment effect. Their framework explains this effect by hypothesizing that the *ambiguity* in the tradeoff for both sellers and buyers is resolved by making effort conservation via automatic System 1 processing the *adoptable* goal. For sellers, the positive attributes of the owned mug are highly accessible in making a judgment, whereas for buyers, the money they will part with is more accessible and adoptable than the mug's attributes. This leads to buyers WTP being lower than sellers WTA. Under conditions that reduce ambiguity (e.g., clear market prices), redirect accessibility (e.g., by focusing buyers on the mug's positive features), or prompt the adoptability of an accuracy enhancement goal via System-2 deliberative processing (e.g., require justification for the judgment), the 3A framework would predict a diminished endowment effect. Thus, without being designed for the endowment effect, the 3A framework can nevertheless organize a range of endowment findings as instances of a more general resource-allocation process in which ambiguity, accessibility, and adoptability interact.

As with the other two strategies, a limitation of the theoretical framework strategy is also apparent. Multiple frameworks can explain the same data, and different criteria might be considered or weighed to determine which is most apt. Nonetheless, these limitations do not undermine the value of any of the three strategies; they instead clarify the scope of the progress claims each can support. Establishing-an-effect work can show that an effect is obtainable under specified conditions, theory-of-the-effect work can identify processes that help explain when it appears or disappears, and theoretical frameworks can locate the effect within broader explanatory systems. The same progression is evident in research on the attraction effect.

Attraction effect

As indicated, the three different strategies can also be observed with the attraction effect. Huber et al.'s (1982) original work exemplifies an *establishing-an-effect* strategy. They document that adding an asymmetrically dominated "decoy" option reliably increases choice of the dominating target, violating regularity, and show that this pattern appears robustly across categories, attribute structures, and designs. The primary contribution is phenomenological: any adequate theory of choice must be able to account for this robust "attraction" pattern, not just explain one-off anomalies.

Subsequent work by Mishra et al. (1993) and Wedell (1991) shifts toward the *theory-of-the-effect* strategy. Mishra, Umesh, and Stem develop a process model in which the decoy alters perceived attribute relevance, similarity, and tradeoffs, and test a causal path model linking variables such as information relevance, product knowledge, and preference strength to the magnitude of the attraction effect. Wedell systematically pits three classes of models against each other in a series of decoy experiments: (1) dimensional-weight models, (2) value-shift models (e.g., range–frequency), and (3) dominance-valuing models. He shows that asymmetric dominance is a necessary and sufficient condition for the effect and that the data are best explained by a dominance-valuing account in which perceiving that the target dominates the decoy directly boosts its global attractiveness, without needing to invoke changes in attribute weights or value scaling.

In Mishra et al. (1993) and Wedell (1991), the attraction effect remains the focal phenomenon, and confidence is claimed by demonstrating that the proposed process predicts when the attraction pattern appears, disappears, or reverses.

Simonson's (1989) reason-based choice account and He and Sternthal's (2023) 3A framework then operate at the level of *theoretical frameworks* that can each explain the attraction effect while also extending far beyond it. Simonson proposes that, under preference uncertainty, consumers often choose the option that is best supported by reasons they can articulate and justify to themselves and to others; the decoy furnishes such a reason for the target, and the same logic extends to other context effects such as compromise choices.

He and Sternthal's (2023) 3A framework reframes the attraction effect not as a special anomaly, but as one instance of how ambiguity in tradeoffs leads people, to make an effort conservation goal adoptable, which activates System-1 processing to rely on whatever comparison standards are most accessible. In a typical attraction set, the ambiguity between a target and a competitor occurs because each dominates on an attribute. Adding a decoy that is similar but inferior to the target facilitates the adoptability of an effort conservation goal that leads to System-1 processing that makes the decoy accessible as a comparison standard. This comparison enhances the choice of the target, that is the attraction effect. By contrast, when System-2 processing is engaged to question adoptability (e.g., recognizing that the decoy is normatively irrelevant), the attraction effect weakens or disappears. In both frameworks, the attraction effect becomes a worked example of a more general explanatory scheme (reason-based choice or 3A), rather than the ultimate object of explanation.

A PROGRESSION IN CLAIMING RESEARCH PROGRESS

As the examples above suggest, we can view the research process in terms of a progression of strategies at different levels of abstraction that fit together to warrant claims of progress synergistically. To emphasize, progression here does not mean that one strategy is superior to another, only that the strategies are at lower to higher levels of abstraction (see Figure 1). Thus, the solution to claiming research progress does not come from a single strategy with its own weaknesses but from the strength of multiple strategies along the progression used together. Each of the three strategies that characterize current practice is insufficient for warranting progress, but they can be used together in a systematic way to justify greater confidence in research progress.

In other words, the three strategies can be viewed as jointly defining what it means to "claim progress." Figure 1 depicts this view of the research process. The contribution of the establishing-an-effect strategy is to identify an effect that is worth studying, one that has to some extent been shown to be replicable, at least when procedures are closely reproduced. With regard to generalization issues, no claim is made that the effect will always be obtained.

The theory-of-the-effect strategy warrants claiming progress when we understand an effect in terms of one or more possible explanatory mechanisms by which the effect could operate. With regard to explanatory depth issues, no claim is made that such explanations provide a unique or complete understanding of the effect.

The theoretical framework strategy provides an explanatory framework that is not tied to specific effects but locates effects in a broader class of behaviors. It connects different effect paradigms and illuminates where new effects might be discovered. Confidence in progress comes from being able to evaluate the theory in comparison to other explanations in terms of relative explanatory power, scope, and other criteria. With regard to underdetermination issues, no claim is made that this is the ultimate true theory.

Thus, while each strategy plays a role, claims of progress should be based on the cumulative outcome of the research process. That is, on the convergence of strategies used in conjunction with the strengths and weaknesses of each considered. Our thesis is that progress can be facilitated by recognizing that different strategies are being used and a more active consideration of their use. This view of progress contrasts markedly with current practice.

Currently, researchers rarely start at the top (the theoretical framework level). Instead, a typical trajectory begins with a striking effect, proceeds to a

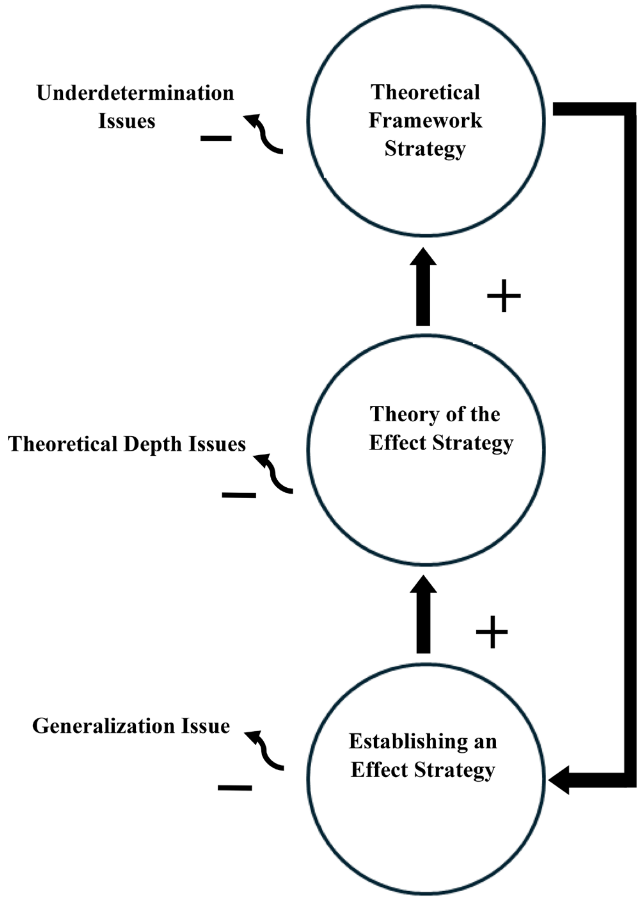

FIGURE 1 System solution for claiming research progress. Figure shows how the three strategies can be related within a broader research process: Establishing effects identifies phenomena, theories of effects specify mechanisms and boundary conditions, and theoretical frameworks organize effects at a higher level of abstraction. The figure also indicates that all three strategies have limitations: Establishing-an-effect work faces generalization issues, theory-of-the-effect work faces theoretical-depth issues, and theoretical framework work faces underdetermination issues.

theory-of-the-effect account of that phenomenon, and only later (if at all) extracts a framework that transcends the original effect paradigm. In practice, however, many projects stop in the middle level: the effect is documented and explained locally, but the opportunity to generalize the explanation into a portable framework is not pursued.

Our aim is not to insist that all work focus on a theoretical framework. Rather, our goal is to encourage more deliberate progression from one strategy to another. In this regard it may be most helpful to consider in the next section progressing from a theory-of-effect strategy to a theoretical framework strategy in more detail.

From theories of effects to theoretical frameworks

How can researchers move from a theory-of-the-effect contribution to a theoretical framework that transcends individual effects? As this is not common in current practice, in this subsection we offer general guidelines and then illustrate them using the 3A framework and several recent theory-of-the-effect papers. We also briefly show how alternative frameworks, such as reason-based choice and evolutionary approaches, can be used to connect and extend these same papers.

General guidelines

Starting from a typical theory-of-the-effect paper, turning it into a framework involves three steps:

1. Abstract the core process beyond the focal effect.
   Ask: What is the general type of situation or tradeoff instantiated by this effect? What variables or constructs are relatively specific to the paradigm (e.g., heritage mugs, cause-related campaigns) and what constructs might capture a more general psychological process (e.g., resource allocation, justification, status motives)?
2. Connect the process to other, structurally similar paradigms.
   Identify existing phenomena that appear to share the same high-level explanation, even if they involve different variables. Ask what the theory would predict in those paradigms and consider designing at least one study that transcends the original effect paradigm to test those predictions.
3. Formulate the contribution in terms of a framework, not only an effect.
   Recast the paper's primary claim as "this is a general account of (broader process, e.g., resource allocation) that can explain and organize the following effects," rather than "this is the explanation of effect E." The original effect then becomes one important *instance* or test case for the framework rather than its primary object.

As indicated, He and Sternthal's (2023) treatment of the attraction effect exemplifies this approach. Rather than treating the attraction effect as the primary endpoint, they use it as one of several paradigms to develop the 3A framework of resource allocation. The key constructs—ambiguity about which goal to prioritize, adoptability of different effort goals, and accessibility of information given the adopted goal—are defined independently of any specific effect. The attraction paradigm then functions as an illustrative example: it shows how the framework can explain both demonstrations and failures to replicate the effect, while the same framework can also be applied to other phenomena such as repetition and depletion.

Tables 3 and 4 summarize how theory-of-the-effect studies can be connected to broader theoretical

TABLE 3 Publications illustrating variation in use of the theory-of-the-effect strategy.

Study	Findings
Christensen and Shu (2024)	Sellers typically demand more for an object than buyers will pay (the endowment effect). Christensen and Shu show that this gap disappears when both parties share a family connection to the object. For example, when grandparents owned a car and both the seller and buyer are relatives, sellers accept lower prices. The authors find that sellers lower their prices because they feel the object's heritage will be preserved rather than lost
DeMotta et al. (2024)	DeMotta, Janssen, and Sen examine how consumers evaluate cause-related marketing campaigns. Analytical thinkers judge campaigns more favorably when a firm's product fits the cause it supports (e.g., a children's clothing brand supporting Save the Children) than when fit is low (e.g., children's clothing supporting Alzheimer's research). Holistic thinkers, by contrast, respond more to how much beneficiaries need help. For instance, they prefer campaigns aiding musicians who suffered accidents over those helping musicians at different career stages. When experimenters prompt analytical thinkers to consider need, they too respond to this information
Boghrati et al. (2023)	Boghrati, Berger, and Packard investigate how writing style affects the success of academic papers. They find that authors who use present tense (“this research shows” rather than “this research showed”), first-person pronouns, and simple articles receive more citations, but only in their introductions, not in methods or results sections. Readers perceive authors who write in present tense as more current and knowledgeable, which drives the effect
Ruan et al. (2024)	Ruan, Polman, and Tanner find that people evaluate experiences more favorably when they are one step from finishing than after they have finished. For example, participants rated a meal kit higher when they had one step left to prepare than when the meal was fully cooked. This preference disappeared when completion felt distant or uncertain, and reversed when participants directly compared the nearly finished and finished states side by side

frameworks. Table 3 first summarizes four recent theory-of-the-effect papers. Table 4 then presents brief accounts of the same findings through three broader theoretical frameworks: the 3A resource-allocation framework, reason-based choice, and a functional-evolutionary framework.

Taken together, Tables 3 and 4 illustrate how theoretical framework thinking can connect theory-of-the-effect studies that, on the surface, address very different topics. At the same time, the tables also underscore the underdetermination issue of the framework strategy: multiple frameworks can often be brought to bear on the same findings.

Accordingly, the purpose of Tables 3 and 4 is not to suggest that any one of the three frameworks has been established as the correct explanation of the studies listed. The tables instead show how framework thinking changes the question asked of theory-of-the-effect papers from “what mechanism explains this effect?” to “what broader explanatory account could connect this effect to other phenomena?” In some cases, multiple frameworks may capture different aspects of a multiply determined phenomenon; in other cases, they may be competing accounts. There is always the potential for new, better frameworks, as well as for not being able to decide between frameworks at a particular point.

One might nonetheless ask: if multiple theoretical frameworks can explain the same data then what do we learn from them? However, one can likewise ask what we learn from the establishing-an-effect strategy as we cannot be assured that just because we have replicated an effect multiple times, we will be able to replicate it another time in another setting. Similarly, we can ask what we learn from the theory-of-the-effect strategy: a moderator that affects whether an effect occurred or did not occur in one setting might be made irrelevant or have its effect reversed in a new setting—and there are always moderators of moderators, and so forth. This is why we argue that while each strategy has limitations, using the strategies synergistically can better warrant claims of progress.

The brief sketches in Table 4 illustrate that once one thinks in terms of frameworks, it becomes natural to ask: How would this effect look from a reason-based perspective? From an evolutionary one? Different frameworks will sometimes yield overlapping accounts; other times they make divergent predictions. Such comparisons can move the field from a collection of unrelated theories-of-effects to a more organized landscape of competing and complementary frameworks. Evaluating these frameworks on criteria such as explanatory power and scope can help us judge the best explanation and inform any theory of the specific effect.

Practical guidelines for researchers

We realize the preceding discussion is a call for reorienting the field's approach to claiming research progress_changes that go beyond the practices of any individual researchers. But the discussion also suggests several actionable steps that an individual researcher can take when designing, conducting, and writing up studies:

1. Be explicit about which strategy you are using. When formulating a project and writing the introduction and discussion, ask: Am I primarily trying to (a) establish that an effect exists, (b) explain when and how a specific effect occurs, or (c) develop or extend a framework that accounts for multiple effects? Making

TABLE 4 Alternative theoretical framework accounts of the studies in Table 3.

Study	3A resource-allocation account	Reason-based choice account	Functional-evolutionary account
Christensen and Shu (2024)	Shared heritage changes how ambiguity about the exchange is resolved: relational preservation becomes more adoptable than gain maximization, and memories or obligations become accessible. The endowment gap narrows because selling within the heritage network can preserve the good's valued meaning	Familial connection changes the reasons sellers and buyers find compelling. Accepting a lower price can be justified by keeping the object in the family or honoring shared history, whereas insisting on a high price may require reasons that seem greedy	Heritage-based valuations can serve kinship and affiliation motives. Maintaining shared possessions signals group identity and loyalty, so the disappearance of the endowment effect reflects protection of social bonds rather than individual resource gain
DeMotta et al. (2024)	Thinking style affects how ambiguity about what to prioritize is resolved and, in turn, which information becomes accessible. Analytical thinking makes firm-cause fit easier to adopt as the relevant standard, whereas holistic thinking makes broader need and context more accessible	Analytical and holistic styles make different justifications compelling. Analytical thinkers can justify a campaign because the partnership "fits"; holistic thinkers can justify it because the beneficiary need is substantial. Moderators that foreground one reason should shift evaluations	Cause-related marketing evaluations may reflect status, affiliation, and moral-signaling motives. Fit and need provide different routes for signaling moral character or group membership; thinking style may alter which signaling opportunity is prioritized
Boghrati et al. (2023)	Clear, confident, and personal language reduces ambiguity about whether an idea is worth cognitive investment and makes favorable interpretive cues, such as currentness and author competence, more accessible. Style matters most in introductions because that is where readers decide whether to allocate deeper processing	Linguistic patterns can supply reasons for endorsing or citing an idea: a direct and confident author gives readers a justification for investing attention, whereas more hedged or impersonal language supplies weaker reasons	Linguistic style may draw on evolved mechanisms for assessing communicator competence and trustworthiness. Clear, confident, personally engaging communication can function as a cue that the communicator is a valuable coalition partner or leader
Ruan et al. (2024)	Being one step from completion makes the goal of finishing highly adoptable and makes the small remaining effort and imminent reward accessible. After completion, effort conservation or goal-reallocation becomes more adoptable; direct comparison changes the accessible standard and reduces the asymmetry	Near completion supplies strong reasons to persist, such as avoiding waste after substantial investment. After completion, the better reasons may favor reallocating effort. Direct comparison encourages more symmetric, normative reasons	Preference for nearly finished tasks may reflect reputational and self-control motives. Failing to complete something almost finished can threaten self-integrity or signal unreliability, whereas completion protects status and the self-image of persistence

this explicit helps align methods, expectations, and claims of progress.

2. Consider the larger research process.
   Locate your current project as to how it can fit into the larger process in Figure 1. Then ask what additional analyses, designs, or reframing would be needed to align with using all three strategies. For an establishing-an-effect-focused paper, this may mean adding theorized mediators and moderators. For a theory-of-the-effect paper, it may mean identifying a higher level of explanation and testing predictions in a second, structurally different paradigm. This recommendation need not require every article to contain a full framework-level empirical package. It can be implemented conceptually by identifying a more abstract level of explanation, describing how the focal effect would be treated under adjacent frameworks, or empirically by adding one study that applies the theorized process outside the original paradigm (see point 4 below).
3. Mine theory-of-the-effect work for framework potential.
   When reviewing your own or others' theory-of-the-effect papers, identify the constructs that are not tied to a specific paradigm and consider how they could be reassembled into a more general account. Ask which other literatures could be brought under the same umbrella (as we did above with the 3A, reason-based, and evolutionary frameworks).
4. Design at least one study that goes beyond the original paradigm.
   A practical way to invoke a framework explanation is to include a study that applies a more abstract explanation of the focal effect in a theory-of-the-effect paper to a different effect paradigm. For example, a paper initially focused on cause-related marketing could add a study testing whether the same theorized resource-allocation or reason-based explanation predicts choices in a non-charitable context.
5. Consider theoretical contributions in terms of frameworks, not just effects.
   In the abstract and discussion, emphasize what the work contributes to a broader explanatory framework

that transcends individual effects: for example, specifying how an evolutionary motive influences diverse behaviors in a modern consumer context. The mechanistic explanation of the focal effect can then be presented as an informative test case rather than as the main contribution.
6. Use alternative frameworks as stress tests. When proposing a new framework, or adopting an existing one, actively consider plausible rival frameworks that can be evaluated in terms of best explanation criteria. Ask what each would predict for your paradigms and where they yield discriminable predictions.

In practical terms, the question is not whether a framework can be invoked after the fact, but whether invoking it provides better insight into a phenomena and broader research area and thereby warrants stronger claims of research progress.

By adopting these practices, researchers can still reap the benefits of rigorous effect demonstration and rich theory-of-the-effect work while redirecting more effort toward building and testing frameworks that organize multiple effect paradigms. In our view, it is this shift from treating effects as endpoints to treating them as inputs into more general theoretical structures that offers the strongest basis for confidence that the field is making enduring progress.

DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available in the supplementary material of this article.

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